Exercise Sheet 12

The notation is that of Lecture 12. The Lie algebra $\mathfrak{g}$ is complex semisimple; modules may be infinite-dimensional. For $\mathfrak{sl}_2$, the Verma module $M(c)$ has basis $v_k = f^k v_0$ ($k \geq 0$), with $v_{-1} = 0$ and

$$hv_k = (c - 2k)v_k, \qquad fv_k = v_{k+1}, \qquad ev_k = k(c - k + 1)v_{k-1}.$$

Exercises 12.1–12.3 develop the lecture. Exercises 12.4–12.6 test where its hypotheses are used.

Exercise 12.1 (Submodules of $\mathfrak{sl}_2$ Verma modules). Let $c \in \mathbb{C}$.

(i) Prove that every submodule W of $M(c)$ is spanned by the vectors v_k that it contains.

(ii) List all submodules of $M(c)$ for $c \notin \mathbb{Z}_{\geq 0}$ and for $c = m \in \mathbb{Z}_{\geq 0}$.

(iii) For $c = m$, prove $N(m) \cong M(-m - 2)$.

Exercise 12.2 (Homomorphisms between Verma modules).

(i) For any $\mathfrak{g}$-module V and $\lambda \in \mathfrak{h}^*$, prove that $\varphi \mapsto \varphi(v_\lambda)$ is a linear isomorphism

$$\mathrm{Hom}_{\mathfrak{g}}(M(\lambda), V) \longrightarrow \{v \in V_\lambda : \mathfrak{n}_+ v = 0\}.$$

(ii) For $\mathfrak{g} = \mathfrak{sl}_2$, compute $\dim \mathrm{Hom}_{\mathfrak{g}}(M(d), M(c))$ for all $c, d \in \mathbb{C}$.

(iii) For $\mathfrak{g} = \mathfrak{sl}_2$, prove that every nonzero module map $M(d) \to M(c)$ is injective.

Exercise 12.3 (Weight spaces of an $\mathfrak{sl}_3$ Verma module). Let $\mathfrak{g} = \mathfrak{sl}_3$ with diagonal $\mathfrak{h}$ and matrix units E_{ij}. Let ε_i read the ith diagonal entry, and take $\alpha_1 = \varepsilon_1 - \varepsilon_2$, $\alpha_2 = \varepsilon_2 - \varepsilon_3$, so that $\mathfrak{n}_+$ is strictly upper triangular. Then $E_{21} \in \mathfrak{g}_{-\alpha_1}$, $E_{32} \in \mathfrak{g}_{-\alpha_2}$ and $E_{31} \in \mathfrak{g}_{-\alpha_1-\alpha_2}$. Take $\beta_1 = \alpha_1$, $\beta_2 = \alpha_2$, $\beta_3 = \alpha_1 + \alpha_2$ and

$$f_1 = E_{21}, \qquad f_2 = E_{32}, \qquad f_3 = E_{31}.$$

(i) Prove $\mathcal{P}(a\alpha_1 + b\alpha_2) = \min(a, b) + 1$ for $a, b \in \mathbb{Z}_{\geq 0}$.

(ii) Give PBW bases of $M(\lambda)_{\lambda - \nu}$ for $\nu = 2\alpha_1 + \alpha_2$ and $\nu = 2\alpha_1 + 2\alpha_2$.

(iii) Write $f_2 f_1 v_\lambda$ in the PBW basis.

Exercise 12.4 (Why the maximal-submodule proof needs weights). Let $\mathfrak{g} = \mathfrak{sl}_2$ and $V = V_0 \oplus V_1$, where $V_0 = \mathbb{C}v$ is trivial and $V_1 = \mathbb{C}^2$ is the defining module with highest vector $w = (1, 0)^{\mathsf{T}}$. Put $u = v + w$. Here V_0 and V_1 name modules of Lecture 6, not weight spaces. As in the lecture, a weight μ of $\mathfrak{sl}_2$ is identified with $\mu(h) \in \mathbb{C}$.

(i) Show that $eu = 0$ and $V = U(\mathfrak{g})u$.

(ii) Find two proper submodules whose sum is V. Conclude that V has no unique maximal proper submodule and has two nonisomorphic simple quotients.

(iii) Which hypothesis of the maximal-submodule theorem fails? Which step of its proof breaks?

Exercise 12.5 (Finite-dimensional highest-weight modules). Let V be a finite-dimensional highest-weight $\mathfrak{g}$-module of highest weight λ.

(i) Use Weyl's theorem from Lecture 5 to prove that V is simple. Deduce $V \cong L(\lambda)$ and $\lambda \in P_+$.

(ii) For $\mathfrak{g} = \mathfrak{sl}_2$, identify every finite-dimensional quotient of $M(c)$, without using Weyl's theorem.

Exercise 12.6 (Endomorphisms without finite dimension). Let $\lambda \in \mathfrak{h}^*$.

(i) Prove $\mathrm{End}_{\mathfrak{g}}(M(\lambda)) = \mathbb{C}\,\mathrm{id}$.

(ii) Prove $\mathrm{End}_{\mathfrak{g}}(L(\lambda)) = \mathbb{C}\,\mathrm{id}$, also when $L(\lambda)$ is infinite-dimensional.

(iii) Why does the proof of Schur's lemma in Lecture 1 not apply directly in (ii)?